Reg. No. | | | | | | | | | | | [14CE751]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2019

AIR POLLUTION AND CONTROL

Time:3 Hours Maximum Marks:70

Instructions:
i. Answer any **FIVE** full Questions choosing at least two from each part.
ii. Draw neat sketches wherever required
iii. Missing data may be suitably assumed.

PART - A

1. a. Give the WHO definition of air pollution and discuss the different sources of air pollution in detail. 08
   b. Explain the effects of air pollution on human health. 06

2. a. List the meteorological parameters that influence the dispersion of pollutants in atmosphere. 04
   b. Define inversion. With neat sketches explain the different types of plume behaviour. 06
   c. Determine the plume rise and effective height of the stack from the following data. Physical height of stack -200 m, inside diameter of stack- 1 m, wind velocity- 3m/s, air temperature 25^0C, barometric pressure- 1000 mb, stack gas velocity- 12m/sec, stack gas temperature- 150^0C. 04

3. a. Enumerate the objectives of air pollution control equipment. 06
   b. Explain with neat sketch working of cyclone separator with advantages and disadvantages. 08

4. a. Explain with neat sketch the operation of electrostatic precipitators with advantages and disadvantages. 08
   b. List and explain types of emission in automobile air pollution. 06

PART - B

5. a. Enumerate the factors on which efficiency of absorption depends. 04
   b. With neat sketch explain packed bed column for removal of gaseous pollutants with advantages and disadvantages. 10

6. a. Define adsorption. List and explain common adsorbents used for removal of gaseous pollutants. 06
   b. Write short notes on; 08
      i. Direct flame incinerator
      ii. Catalytic incinerator

7. a. Write the difference between adsorption and absorption with examples. 06
   b. With neat sketch explain the working of bio-filters with advantages and disadvantages. 08

8. a. Define noise pollution. List the sources and effects of noise pollution. 08
   b. Enumerate noise pollution control methods. 06